Confluent Kafka Isotope

`confluent-kafka-isotope` is a reference implementation of an ***e-commerce order pipeline that uses Kafka Interceptors, Prometheus, and Apache Flink to capture, analyze, and report end-to-end event-tracing data in both batch and near real time***.

Much like isotopes used to trace molecules through a biochemical pathway, each event carries lightweight metadata that allows it to be followed as it moves through Kafka topics and distributed microservices.

Kafka topics become the connective tissue between services, while Kafka Interceptors quietly transform the event pipeline itself into an observable distributed system.

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1.0 How an Isotope Traverses an Event Pipeline

An **Isotope** is a ***lightweight tracing artifact attached to Kafka record headers***. Like a biochemical isotope used to trace molecules through a metabolic pathway, it allows the journey of a record through an event-driven architecture to be observed and analyzed.

This project models a Kafka pipeline as a **bipartite graph**: *services occupy one vertex set*, *topics the other*, and *every produce and consume operation forms an edge between them*. The resulting graph provides a unified view of the complete event topology, from producers to intermediate processing services to terminal consumers.

A `ProducerInterceptor` stamps each record with an isotope and appends one hop for every `send()` operation, capturing the produce edges. Consumers call `IsotopeContext.recordConsume()` to emit a lightweight marker representing the corresponding consume edges, while services that consume and then produce invoke `IsotopeContext.adoptFromRecord()` so the trace identity persists across every hop. Apache Flink reconstructs the complete **service → topic → service** graph from the isotope headers alone, producing seven reports: end-to-end latency, latency percentiles, produce-side topology, the full bipartite topology, hop distribution, per-topic coverage (a trace-loss funnel signal), and stuck-trace

detection. The same implementation runs unchanged on both Confluent Platform (self-managed) with Confluent Manager for Apache Flink (CMF) and Confluent Cloud for Apache Flink (CCAF).

> **Full architectural design of Isotope Tracing** — the header layout (`x-isotope` JSON + seven scalar headers with a worked example), how the producer interceptor gets invoked, why the consume side uses explicit calls instead of a `ConsumerInterceptor`, and the bipartite-graph rationale — is in **docs/design.md**.

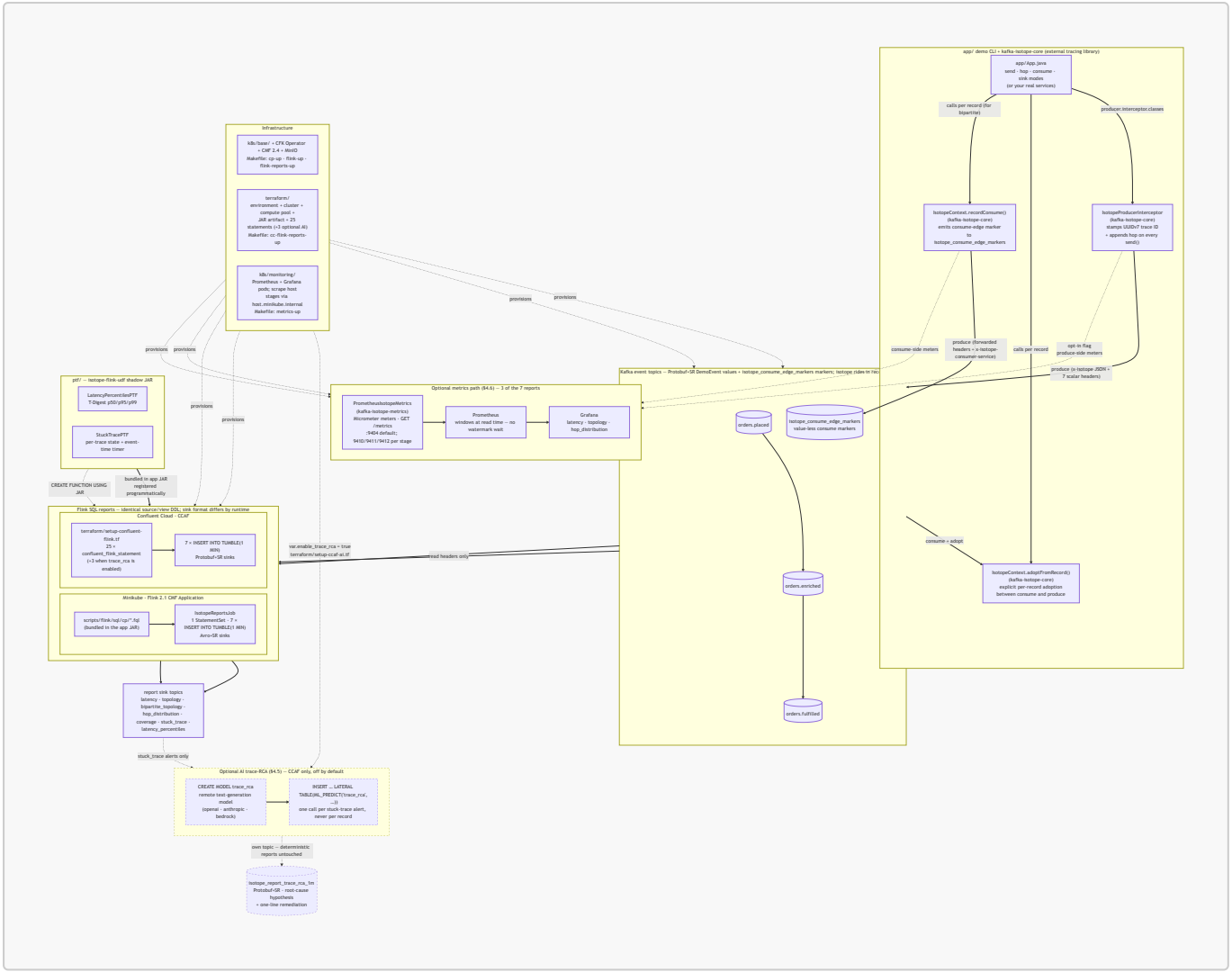
2.0 Project's Architecture

A bird's-eye view of the moving parts. The demo CLI in `app/` consumes the external tracing library (`ai.signalroom:kafka-isotope-core`), which registers a Kafka `ProducerInterceptor` that stamps an isotope into record headers on every `send()`. Consume-then-produce services propagate the inbound trace by explicitly calling `IsotopeContext.adoptFromRecord(record)`. Business events then flow through a three-topic Kafka pipeline, where Flink SQL reads the isotope metadata and emits one-minute aggregate reports.

Both runtimes run the same seven logical reports off the same source and view definitions, though each runtime keeps its own copy of the SQL. **Confluent Platform (CP)** on Minikube executes the `.fql` files under `scripts/flink/sql/cp/` as a single Flink 2.1 Confluent Manager for Apache Flink (CMF) Application (`IsotopeReportsJob`), while **Confluent Cloud for Apache Flink (CCAF)** applies the same logical SQL as inline `confluent_flink_statement` Terraform resources under `terraform/`. The shadow JAR from `ptf/`—which powers two of the seven reports—runs unchanged on both runtimes: bundled into the CP application JAR and uploaded as a Flink artifact on CCAF.

Alongside that—**additive, opt-in, and disabled by default**—the interceptor can also emit **Micrometer** metrics for **Prometheus**, with **Grafana** providing visualization (§4.6). This path produces the three **stateless** reports (`latency_1m`, `topology_1m`, and `hop_distribution_1m`) without requiring a stream processor. The remaining four reports continue to run in Flink because they depend on per-`trace_id` state or absence-of-event analysis, which falls outside Prometheus's query model.

(Kafka is drawn once below for brevity; each runtime provisions its own Kafka cluster.)



See §3.0 for the file tree behind each box, and §4.0 for the run commands.

3.0 Project layout

The isotope tracing library lives in its own repo — **j3-signalroom/kafka-isotope** (`ai.signalroom:kafka-isotope-core` + `ai.signalroom:kafka-isotope-metrics`). This repo is the runnable demo that consumes it.

```
app/                                        demo CLI + tests (consumes the
isotope library)
  src/main/proto/ai/signalroom/kafka/isotope/proto/
    demo_event.proto                        DemoEvent message (Protobuf value
schema)
  src/main/java/ai/signalroom/kafka/isotope/
    App.java                                demo CLI — pipeline-position verbs
                                            (place / enrich / fulfill / ship) +
generic
                                            send / hop / consume / sink modes;
registers
                                            IsotopeProducerInterceptor + starts
the
                                            PrometheusIsotopeMetrics exporter
(§4.6)
```

```
  src/integrationTest/java/.../         BrokerSmokeIT,
ProducerInterceptorIT,
                                        ThreeStageHopPropagationIT,
BipartiteTopologyIT,
                                        IsotopeTestHarness — live-broker
tests; produce/consume
                                        DemoEvent via SR-framed Protobuf
                                        (need Minikube CP + SR port-
forwarded)
ptf/                                    Flink reports application + PTF
shadow JAR
  src/main/java/ai/signalroom/kafka/isotope/flink/
    IsotopeReportsJob.java              CP entry point — reads the bundled
sql/*.fql,
                                        registers both PTFs
programmatically, runs the
                                        7 INSERT INTOs as one StatementSet
(CMF Application)
    LatencyPercentilesPTF.java          T-Digest p50/p95/p99 (PTF: per-
window state + timers)
    StuckTracePTF.java                  per-trace state + event-time timer
    TDigests.java                       shared T-Digest (de)serialization
  src/test/java/.../                    TDigestsTest
k8s/base/                               CFK / CMF manifests (applied by
`make cp-up` / `flink-up`)
  confluent-platform-c3++.yaml          Kafka / SR / Connect / ksqlDB /
Control Center
  minio.yaml                            in-cluster S3-compatible store
backing CMF's
                                        cmf:// artifact (JAR) storage
  cmf-values.yaml                       Helm values for CMF 2.4 — artifact
storage
                                        (points at MinIO) + writable
environment catalog
  cmf-flink-application.json            FlinkApplication template
(envsubst'd by
                                        deploy-cmf-flink-reports.sh) — the
reports job
  flink-sql-isotope.Dockerfile          custom cp-flink image for the CMF
compute pool:
                                        bakes in the Kafka + avro-confluent
SQL connectors
                                        and the s3-fs-hadoop plugin (`make
flink-image-build`)
  flink-cluster-deployment.yaml         optional cp-flink session cluster
for ad-hoc
                                        `make flink-deploy` / `make flink-
sql`
                                        (not used by the reports path)
  flink-rbac.yaml                       RBAC for the cp-flink operator
k8s/monitoring/                         optional metrics showcase (§4.6) —
`make metrics-up`
  00-namespace.yaml                     dedicated 'monitoring' namespace
  10-prometheus.yaml                    Prometheus pod/Service; scrapes
```

```
host stages
                                          via
host.minikube.internal:9410/9411/9412
  20-grafana.yaml                         Grafana pod/Service; auto-
provisioned datasource
                                          + 8-panel dashboard for the 6
produce/consume meters
  kustomization.yaml                      `kubectl apply -k k8s/monitoring`
  README.md                               runbook + troubleshooting
scripts/
  port-forward-kafka.sh                   localhost:30092 → Kafka,
localhost:8081 → SR
  port-forward-taskmanager.sh             Flink TaskManager web UI forward
  deploy-cmf-flink-reports.sh             builds shadow JAR + uploads it as a
cmf://
                                          artifact + deploys the reports as a
Flink 2.1
                                          CMF Application (runs sql/cp/*.fql)
  deploy-cc-flink-reports.sh              builds shadow JAR + wraps
`terraform apply`
                                          for the CCAF path
  cc-cli-env.sh                           pulls Kafka + SR creds from
`terraform output`,
                                          builds the JAAS string, exports
BOOTSTRAP /
                                          SR_URL / KAFKA_KEY / KAFKA_SECRET /
JAAS / ...
  cc-app-run.sh                           thin wrapper around `./gradlew
:app:run` that
                                          sources cc-cli-env.sh and injects
the six -D flags
  flink/README.md                         Flink SQL reports — runtime split
(CP=7 reports/Avro+SR,
                                          CCAF=7 reports/Protobuf+SR),
layout, operations
  flink/sql/cp/                           CP Flink SQL: 00_source_table,
01_register_functions,
                                          05_isotope_view,
06_consume_events_view,
                                          05_report_sinks (avro-confluent),
                                          10/20/25/30/40/60/70 INSERT INTO
reports, 99_teardown
                                          (CCAF SQL is inlined under
terraform/setup-confluent-flink.tf.)
  flink/sql/ccaf-ai/                       optional AI extension (CCAF-only,
off by default)
    trace_rca.fql                         PoC: 8th, AI-generated report —
turns each
                                          stuck-trace alert into an LLM root-
cause
                                          hypothesis via CREATE MODEL +
ML_PREDICT
                                          (wired in by terraform/setup-ccaf-
ai.tf)
```

```
terraform/                                CCAF infrastructure-as-code (`make
cc-flink-reports-up`)
  providers.tf                            Confluent provider — cloud
key/secret vars
  versions.tf                             required Terraform (>= 1.13) +
provider versions
  variables.tf                            confluent_api_key/secret, cloud,
region, day_count
  data.tf                                 organization lookup + other data
sources
  setup-confluent-environment.tf          environment (ESSENTIALS stream-
governance package)
  setup-confluent-kafka.tf                Kafka cluster + Kafka API key
rotation module
                                          (iac-confluent-api_key_rotation-
tf_module)
  setup-confluent-flink.tf                service account + 6 role bindings,
compute pool,
                                          artifact upload, SR API key
rotation, and 25 inline
                                          `confluent_flink_statement`
resources: 4 ALTER TABLE
                                          + 3 VIEW + 7 sink CREATE TABLE + 2
DROP FUNCTION +
                                          2 CREATE FUNCTION (both PTFs) + 7
INSERT INTO
  setup-ccaf-ai.tf                        OPTIONAL AI trace-RCA report — 3
extra
                                          statements (CREATE MODEL + Protobuf
sink +
                                          INSERT … ML_PREDICT); gated on
                                          var.enable_trace_rca (default
false)
  outputs.tf                              environment_id, bootstrap, SR URL,
rotating
                                          Kafka + SR API key/secret outputs
(sensitive)
docs/                                     extracted long-form docs (linked
from the README)
  design.md                               isotope tracing deep-dive
  gen_isotope_diagram.py                  script to generate the isotope
architecture diagram
  isotope_diagram.png                     visual representation of the
isotope and its flow through Kafka headers
  runbook-minikube.md                     full CP-on-Minikube run sequence
(§4.4)
  runbook-ccaf.md                         full CCAF / Terraform run sequence
(§4.5)
  metrics.md                              Micrometer/Prometheus meter +
PromQL reference (§4.6)
  terraform.png                           rendered resource graph (embedded
in §4.5)
Makefile                                  cp-up / flink-up / kafka-pf-up /
flink-reports-up /
```

```
                                                cc-flink-reports-up / cc-flink-
reports-down /
                                                metrics-up / metrics-down /
metrics-delete / ...
```

4.0 Getting Started

First, clone the repo to your local machine using the GitHub CLI:

```
gh repo clone j3-signalroom/confluent-kafka-isotope
```

Change to the repo directory:

```
cd /path/to/confluent-kafka-isotope
```

Then decide what you want to run:

4.1 Unit tests (no broker, instant)

```
./gradlew test                          # runs :ptf:test (the app has no unit
tests in this repo)
# or scoped:
./gradlew :ptf:test                     # TDigestsTest — T-Digest sketch
(de)serialization + accuracy
```

4.2 Demo CLI — see one trace propagate live

The fastest way to watch the isotope mechanic. Requires the cluster to be up and the Kafka + SR forwards running (see step 3 below for the bring-up commands). The CLI has two argument styles — **pipeline-position verbs** that bake in the orders.* topic chain (recommended for the demo) and **generic verbs** that take raw topic + service args (for ad-hoc inspection on any topic):

Verb	Args	What it does
`place`	`[payload]`	Produces one isotope-tagged `DemoEvent` to `orders.placed` as `order-intake-service` (default payload: `hello`), then exits. Auto-creates the topic.
`enrich`	—	Consumes from `orders.placed`, adopts the isotope, **emits a consume-edge marker to `isotope_consume_edge_markers` as `order-enrichment-service`**, then re-produces to `orders.enriched`. Runs until Ctrl-C.

Verb	Args	What it does
`fulfill`	—	Same as `enrich` but for `orders.enriched → orders.fulfilled` as `order-fulfillment-service`. Runs until Ctrl-C.
`ship`	—	Terminal consumer for `orders.fulfilled` as `shipping-notification-service`. **Emits a consume-edge marker** so it shows up in the bipartite report; does not re-produce. Runs until Ctrl-C.
`send`	`<topic>` `<service>` `<payload>`	Generic produce. Auto-creates the topic.
`hop`	`<in-topic>` `<out-topic>` `<service>`	Generic consume-then-produce; emits a consume-edge marker. Runs until Ctrl-C.
`consume`	`<topic>` `<service>`	Generic terminal-consume; emits a consume-edge marker and pretty-prints the trail. Runs until Ctrl-C.
`sink`	`<topic>`	Passive peek — pretty-prints the isotope trail but does NOT emit a consume marker. Use for ad-hoc inspection. Runs until Ctrl-C.

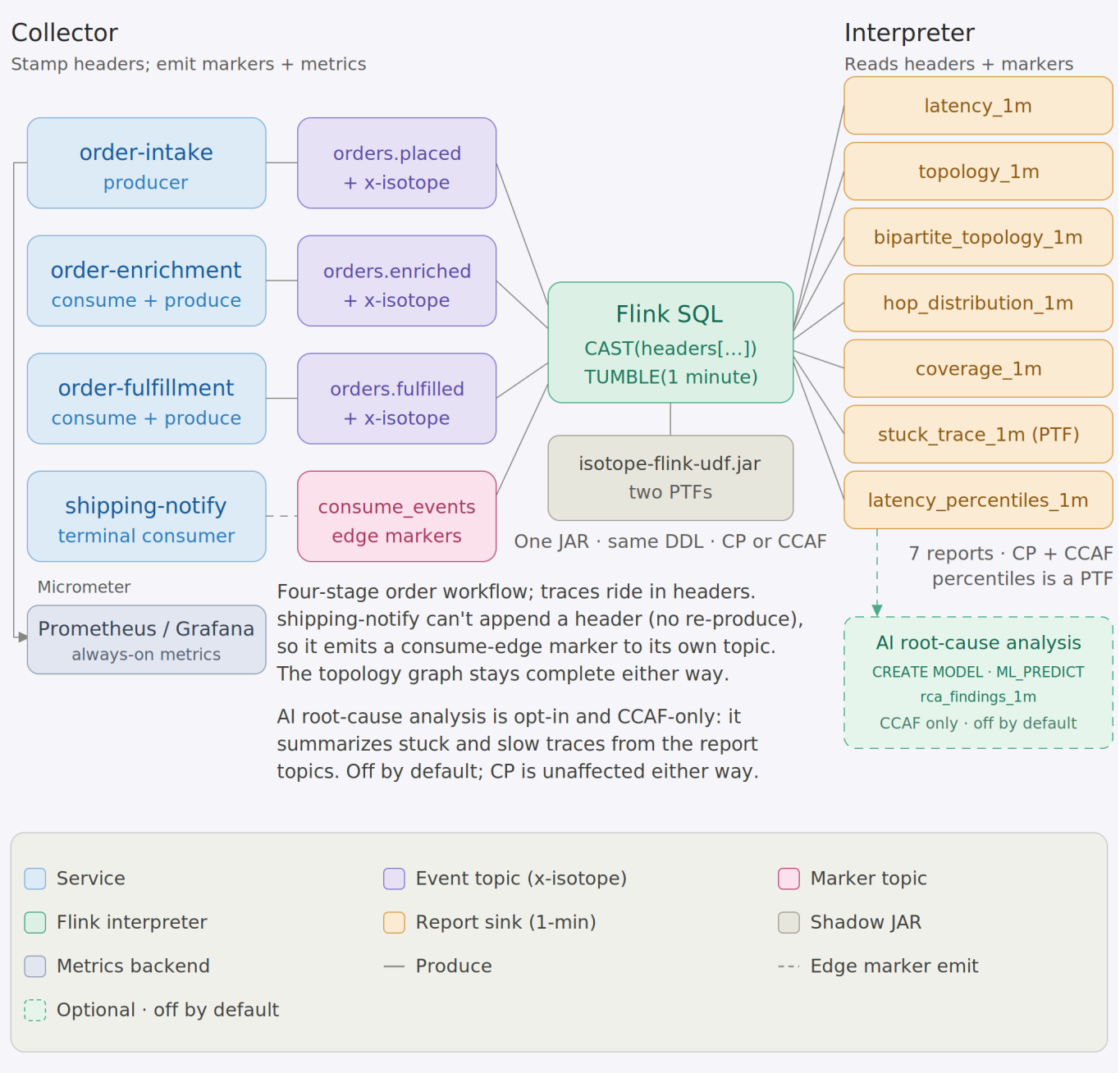
4.2.1 Full Bipartite Demo

A 4-stage chain (full bipartite graph) in four terminals. Run them in pipeline order — `place` produces, then `enrich` / `fulfill` / `ship` each pick up where the previous stage left off:

```
# Terminal A — kick the chain off (run repeatedly to send more)
./gradlew :app:run --args="place 'hello world'" -q

# Terminal B — first hop: orders.placed → orders.enriched
./gradlew :app:run --args="enrich" -q

# Terminal C — middle hop: orders.enriched → orders.fulfilled
./gradlew :app:run --args="fulfill" -q

# Terminal D — terminal consumer (prints the full 3-hop trail AND emits a
#              consume-edge marker so shipping-notification-service shows
up
#              in the bipartite report)
./gradlew :app:run --args="ship" -q
```

Terminal D's output for each record shows the same `trace_id` across all three hops, `origin = order-intake-service` (never reassigned), and `hops[]` listing `order-intake-service → order-enrichment-service → order-fulfillment-service` in order with per-hop timestamps. The bipartite-topology report sees all six edges: produce edges `order-intake-service → orders.placed`, `order-enrichment-service → orders.enriched`, `order-fulfillment-service → orders.fulfilled` and consume edges `orders.placed → order-enrichment-`

`service`, `orders.enriched → order-fulfillment-service`, `orders.fulfilled → shipping-notification-service`. Swap `consume` for `sink` if you only want to inspect records without recording the terminal edge. Override endpoints via `-Dkafka.bootstrap=…` / `-Dschema.registry.url=…` if you're not on the default Minikube layout.



> **Just want the commands?** The full local stack-up sequence (cluster → Kafka → Flink → reports → traffic → teardown) is consolidated in **docs/runbook-minikube.md**.

4.3 Integration tests (live Kafka via Minikube)

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start                    # one-time
make cp-up                             # CFK Operator +
Kafka/SR/Connect/ksqlDB/C3 (~5 min)
```

```
make kafka-pf-up                    # localhost:30092 → Kafka,
localhost:8081 → Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest                                    #
every IT below
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT'   #
just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
    -PkafkaBootstrap=localhost:30092 \
    -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
`BrokerSmokeIT`	AdminClient can create/list/delete a topic via the NodePort port-forward
`ProducerInterceptorIT`	A consumer sees the `x-isotope` JSON header + all 7 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves `DemoEvent.source` / `payload`
`ThreeStageHopPropagationIT`	`order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service` produces a stable trace ID, 2-hop trail in send order, and correct scalar headers (origin = `order-intake-service`, this = `order-enrichment-service`, hop count = 2) at the terminal; consume-then-produce hops use `IsotopeContext.adoptFromRecord` to carry the trace forward
`BipartiteTopologyIT`	The 4-stage `order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service → topic-CD → shipping-notification-service` chain emits exactly three consume-edge markers to a per-test markers topic — one per consume edge. Every marker carries the trace ID, forwarded `x-isotope-*` scalars describing the

Test	What it verifies
	upstream producer, and the new `x-isotope-consumer-service` naming the downstream consumer. Asserts the `(consumer_service, consumed_topic)` set is exactly the three pairs of stages 2-4

4.4 Confluent Platform on Minikube — Production-Like Streaming, Running Locally

> **Caveat:** Minikube is a single-node cluster. It is not a production-like environment, but it is sufficient for local development and testing. The Confluent Platform (CP) stack runs in Minikube, and you can port-forward Kafka and Schema Registry to your local machine.

To ***run***, ***test***, and ***debug*** Apache Flink like a production engineer, this project provides a full Confluent Platform stack running locally on Minikube — no cloud required.

You get a production-like environment on your machine, with all the components you'd expect in a real deployment:

- **Confluent Platform** (KRaft mode) via Confluent for Kubernetes (CFK)
- **Apache Flink 2.1.2** via the Confluent Flink Kubernetes Operator 1.140.1
- **Confluent Manager for Apache Flink (CMF) 2.4.0** for Flink environment management

To run this project, you'll need **macOS (with Homebrew)** or **Linux (with apt-get)**. The full stack — **Minikube + Confluent Platform + Flink + CMF** — is resource-intensive and designed to mirror a production environment. Therefore, the following defaults are recommended:

Resource	Default
CPUs	6
Memory	20 GB
Disk	50 GB

> These settings ensure stable performance across all components. You can tune them as needed, but lower resource levels may cause pod restarts or degraded performance.

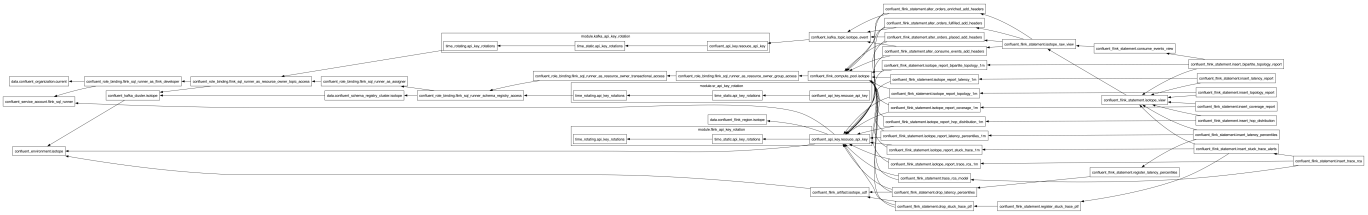
Seven reports — five pure Flink SQL plus two JAR-backed PTFs — run as a **single Flink 2.1 CMF Application** (`IsotopeReportsJob`, one StatementSet, seven sinks) submitted through CMF and executed by the Confluent Flink Kubernetes Operator. No raw session cluster is involved; `k8s/base/flink-cluster-deployment.yaml` (`make flink-deploy` / `make flink-sql`) is a separate, optional path for ad-hoc SQL. Confluent Cloud runs the same seven reports from its own copy of the SQL, inlined in Terraform — see **§4.5** for that path; this section is the local-Minikube one.

The full bring-up sequence — cluster → Flink → reports → traffic → teardown — is consolidated in docs/runbook-minikube.md. The short version: `make flink-up` then `make flink-reports-up`, then drive traffic across **multiple** 1-minute windows (a single burst sits in one open window forever — the watermark has to cross `window_end` for a tumbling window to emit) and wait ~90s after the last record.

Report sink topics ride **Avro+SR** (`avro-confluent`, auto-registered on first write) so Control Center renders them natively — a deliberate *format-by-domain* split: app events are **Protobuf+SR** (`DemoEvent`), Flink aggregates are **Avro+SR** (cp-flink ships no SR-integrated Protobuf format), and the consume-edge marker topic `isotope_consume_edge_markers` is **null-value / headers-only**. Not a defect — a clean split by domain.

4.5 Flink SQL reports on Confluent Cloud for Apache Flink (CCAF)

The CCAF parallel of §4.4, driven by Terraform under terraform/ — no local cluster. One `make` target spins up a fresh Confluent Cloud environment, Kafka cluster, compute pool, two rotating service-account API key pairs (Kafka + Schema Registry), the uploaded PTF JAR, and 25 `confluent_flink_statement` resources (4 ALTER TABLE + 3 VIEW + 7 sink CREATE TABLE + 2 CREATE FUNCTION + 7 INSERT INTO, plus 2 transient DROP FUNCTION). The shape mirrors `apache_flink-kickstarter-ii` — same provider version and `iac-confluent-api_key_rotation-tf_module`.



The full provision → deploy → traffic → teardown sequence is consolidated in docs/runbook-ccaf.md. The short version: `export` a Cloud-level Confluent Cloud API key, `make cc-flink-reports-up` (~6–8 min first run; idempotent re-applies), drive traffic with `scripts/cc-app-run.sh place|enrich|fulfill|ship` across **multiple** 1-minute windows (a single burst sits in one open window forever, and you wait ~90s after the last record), then `make cc-flink-reports-down` to `terraform destroy` the whole environment.

Format-by-runtime (not-by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in scripts/flink/sql/cp/05_report_sinks.fql). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's WITH clause in terraform/setup-confluent-flink.tf). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in scripts/flink/sql/cp/, CCAF's lives inline as `confluent_flink_statement` resources in terraform/setup-confluent-flink.tf.

Why percentiles is a PTF. CCAF rejects all `CREATE FUNCTION` statements for user-defined *aggregate* functions ("aggregate functions are not supported"). Percentiles would naturally be an aggregate, so to keep the report portable it's implemented as a `ProcessTableFunction` instead — `LATENCY_PERCENTILES` (class `LatencyPercentilesPTF`) does its own 1-minute tumbling-window aggregation over a T-Digest sketch via per-window state and event-time timers. A PTF has no such restriction, so it registers and runs on **both** runtimes, exactly like `STUCK_TRACE_PTF`. Both runtimes therefore run the same seven reports: `latency` (avg/min/max), `topology` (produce-side), `bipartite_topology` (full service↔topic↔service graph), `hop_distribution`, `coverage`, `stuck_trace`, and `latency_percentiles` (p50/p95/p99).

Optional: AI root-cause analysis (CCAF-only, off by default). Beyond the seven deterministic reports, terraform/setup-ccaf-ai.tf wires an **eighth, AI-generated report** that turns each stuck-trace *alert* into a natural-language root-cause hypothesis plus a one-line remediation. It adds three `confluent_flink_statement` resources — `CREATE MODEL trace_rca` (a remote text-generation model), a `proto-registry` sink `isotope_report_trace_rca_1m`, and an `INSERT … SELECT …`

`LATERAL TABLE(ML_PREDICT('trace_rca', …))` — all **gated on `var.enable_trace_rca` (default `false`)**, so a normal `make cc-flink-reports-up` is completely un``affected. The model is invoked once per *alert* (the low-volume 1-minute stuck-trace report topic), never per record, and its output lands on its own topic — it never overwrites the deterministic reports. Defaults target OpenAI (`gpt-4o`); for Claude hosted by Anthropic — a **direct** CCAF provider — set `rca_model_provider = "anthropic"`, `rca_model_endpoint = "https://api.anthropic.com/v1/messages"`, a Claude `rca_model_version`, and your Claude `rca_model_api_key` (bare API key, no AWS auth), plus the Anthropic-required `rca_model_max_tokens`. Claude via AWS Bedrock (`rca_model_provider = "bedrock"`) also works but isn't required. The standalone SQL PoC — a `CREATE MODEL` + `ML_PREDICT` walkthrough with provider notes and alternative options — is in scripts/flink/sql/ccaf-ai/trace_rca.fql.

4.6 Stateless reports via Micrometer → Prometheus/Grafana (optional)

Three of the seven reports — `latency_1m`, `topology_1m`, `hop_distribution_1m` — are pure stateless scalar aggregation over bounded-cardinality dimensions (service / topic / hop_count, never `trace_id`). Those don't need a stream processor: the producer interceptor already has every value in scope on each `send()`, so it emits them as **Micrometer** meters (PrometheusIsotopeMetrics.java) and lets **Prometheus** window them at read time, with **Grafana** on top. The other four (`latency_percentiles`, `coverage`, `bipartite_topology`, `stuck_trace`) are per-`trace_id` stateful or absence-of-event problems Prometheus can't express, so they **stay in Flink**. This path is **additive and opt-in** — a 3-Micrometer / 4-Flink split.

> **Full details** — the meter/PromQL reference, the produce- and consume-side signals, the two deliberate gaps (`distinct_traces`, windowed `min`), and why percentiles stays a PTF — are in **docs/metrics.md**. The one-command Prometheus + Grafana showcase has its own runbook: **k8s/monitoring/README.md** (`make metrics-up`).

5.0 Resources

- Medium Article: Kafka's quiet observability superpower — Kafka Interceptors